

Lab 3 - Routing Protocol

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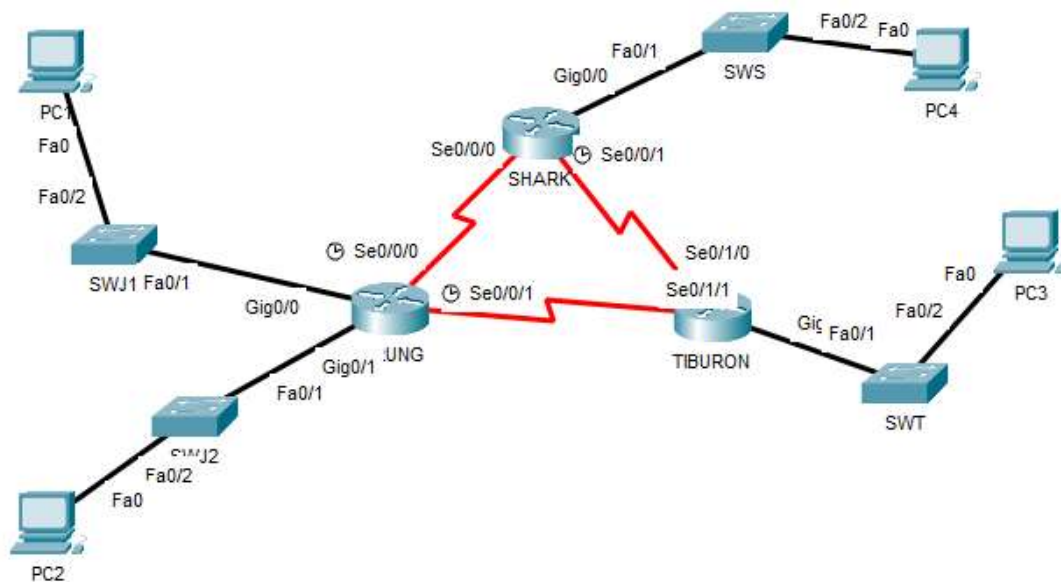
MATRIC NUM.: A23CS0236

SECTION: 02

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.1	255.255.255.224	-
4		G0/1	172.18.110.33	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.129	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.65	255.255.255.192	-
11	PC1	-	172.18.110.30	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.62	255.255.255.224	172.18.110.33
13	PC3	-	172.18.110.190	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.126	255.255.255.192	172.18.110.65

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
2. Using the IP addresses calculated, configure the devices with the appropriate information.

Network address given: 172.18.110.0/23 (network portion – 23 bit, host portion – 9 bit)

Total host = $2^9 = 512$ host

IP address range = 172.18.110.0 – 172.18.111.255

Usable IP address = 172.18.110.1 – 172.18.111.254

Device	Subnetwork	Usable Hosts	Host Portion (bit)	Prefix	Subnet Mask (Decimal)
JERUNG	LAN1	20	$2^5 = 32$ (5 bit)	/27	255.255.255.224
	LAN2	30	$2^5 = 32$ (5 bit)	/27	255.255.255.224
SHARK	LAN	60	$2^6 = 64$ (6 bit)	/26	255.255.255.192
TIBURON	LAN	50	$2^6 = 64$ (6 bit)	/26	255.255.255.192
Connections	JERUNG-SHARK	2	$2^2 = 4$ (2 bit)	/30	255.255.255.252
	JERUNG-TIBURON	2	$2^2 = 4$ (2 bit)	/30	255.255.255.252
	SHARK-TIBURON	2	$2^2 = 4$ (2 bit)	/30	255.255.255.252

Device	Subnetwork	IP address boundaries	Usable IP address
JERUNG	LAN1	172.18.110.0 – 172.18.110.31	172.18.110.1 – 172.18.110.30
	LAN2	172.18.110.32 – 172.18.110.63	172.18.110.33 – 172.18.110.62
SHARK	LAN	172.18.110.64 – 172.18.110.127	172.18.110.65 – 172.18.110.126
TIBURON	LAN	172.18.110.128 – 172.18.110.191	172.18.110.129 – 172.18.110.190
Connections	JERUNG-SHARK	172.18.110.192 – 172.18.110.195	172.18.110.193 – 172.18.110.194
	JERUNG-TIBURON	172.18.110.196 – 171.18.110.199	172.18.110.197 – 171.18.110.198
	SHARK-TIBURON	172.18.110.200 – 172.18.110.203	172.18.110.201 – 172.18.110.202

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
 - a. CLI
 - i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```

SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0

```

Figure A

JERUNG

```

show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
L       172.18.110.33/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1

```

SHARK

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1
```

TIBURON

```
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.128/27 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0
```

b. PT tool

- i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

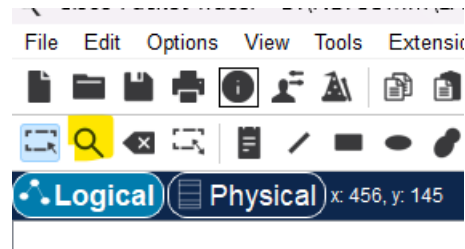


Figure B

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

SHARK

Routing Table for SHARK				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

TIBURON

Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

2. Try to ping PC2 from PC1, paste the results here.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.

- PC1 and PC4 are in different subnets, meaning they belong to separate network segments.
- Devices in different subnets cannot communicate directly because their traffic needs to be routed through a router that knows how to forward packets between the subnets.

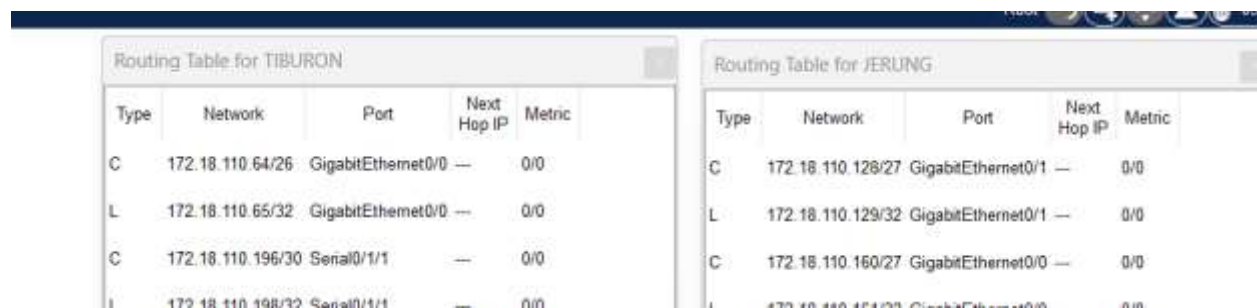
5. What needs to be done to ensure all PCs can ping each other successfully?

- Configure routers to enable communication between different subnets. This can be done using dynamic routing protocols like RIP, OSPF, or static routes.

- Assign the correct default gateway to each PC, pointing to the IP address of the router interface connected to its subnet.

Task 2 – Routing Configuration

1. Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.



Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	—	0/0
L	172.18.110.65/32	GigabitEthernet0/0	—	0/0
C	172.18.110.196/30	Serial0/1/1	—	0/0
L	172.18.110.198/32	Serial0/1/1	—	0/0

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	—	0/0
L	172.18.110.129/32	GigabitEthernet0/1	—	0/0
C	172.18.110.160/27	GigabitEthernet0/0	—	0/0
L	172.18.110.161/32	GigabitEthernet0/0	—	0/0

Figure D

2. In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- a. What can you say about the addresses used in the 'network' instructions in Figure E?

The addresses used in the network instructions in Figure E represent the subnet IDs of the directly connected networks to the router JERUNG.

172.18.110.128 - This is a subnet ID representing one of the networks directly connected to Router JERUNG.

172.18.110.160 - Another subnet ID connected to JERUNG, likely representing a different LAN or another router's connection.

172.18.110.192 - This represents a point-to-point link subnet between Router JERUNG and another device (e.g., Router TIBURON or SHARK).

172.18.110.196 - This is also a point-to-point link subnet, possibly another connection to a different router or device.

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Routing Table for JERUNG					Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0	R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0	C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0	L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/27	Serial0/0/1	172.18.110.198	120/1	R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0	C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0	L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0	C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0	L	172.18.110.202/32	Serial0/1/0	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1					

- a. What are the changes seen in TIBURON?

- TIBURON starts advertising the subnets 172.18.110.64, 172.18.110.196, and 172.18.110.200 to neighboring routers.
- TIBURON's routing table is updated to include routes learned via RIP from neighboring routers, allow it to reach other subnets in the network.

b. What are the Networks with type R in TIBURON and JERUNG?

- TIBURON – 172.18.110.0/27, 172.18.110.32/27, 172.18.119.192/30
 - JERUNG – 172.18.110.128/26, 172.18.110.200/30

c. Ping PC3 from PC1. Was it successful?

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.190: bytes=32 time=9ms TTL=126
Reply from 172.18.110.190: bytes=32 time=5ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 9ms, Average = 5ms
```

Success

d. Ping PC4 from PC1. Was it successful?

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Unsuccess

e. Explain the reasons for your answer.

- PC1 and PC3 success because they are in same subnets that directly connected to JERUNG. JERUNG can route between its interface.

- PC1 and PC4 fail because they are in different subnets that connected to different router. PC1 connected to JERUNG but PC4 connected to SHARK.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.128
SHARK(config-router)#network 172.18.110.160
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.196
SHARK(config-router)#no auto-summary
SHARK(config-router)#
SHARK(config-router)#
```

- g. Open router SHARK's routing table and paste here.

Routing Table for SHARK				
Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	<pre> C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=9ms TTL=126 Reply from 172.18.110.30: bytes=32 time=8ms TTL=126 Reply from 172.18.110.30: bytes=32 time=8ms TTL=126 Reply from 172.18.110.30: bytes=32 time=10ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 8ms, Maximum = 10ms, Average = 8ms </pre>
2	<pre> C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Request timed out. Reply from 172.18.110.62: bytes=32 time=10ms TTL=126 Reply from 172.18.110.62: bytes=32 time=3ms TTL=126 Reply from 172.18.110.62: bytes=32 time=8ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 3ms, Maximum = 10ms, Average = 7ms </pre>
3	<pre> C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=10ms TTL=126 Reply from 172.18.110.190: bytes=32 time=9ms TTL=126 Reply from 172.18.110.190: bytes=32 time=4ms TTL=126 Reply from 172.18.110.190: bytes=32 time=7ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 4ms, Maximum = 10ms, Average = 7ms </pre>

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: [192.168.1.0](#)
 - b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.254
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

- c. After this change, can PC4 and PC1 ping PC3?

```
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

[PC1 to PC3 FAIL](#)

```
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.
Request timed out.

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

[PC4 to PC3 FAIL](#)

d. Copy and paste the routing tables for both SHARK and TIBURON here.

Routing Table for SHARK					Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0	R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
L	172.18.110.194/32	Serial0/0/0	---	0/0	C	172.18.110.196/30	Serial0/1/1	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	L	172.18.110.198/32	Serial0/1/1	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	C	172.18.110.200/30	Serial0/1/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0	L	172.18.110.202/32	Serial0/1/0	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0	C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
					L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

e. Referring to the routing table, explain your findings.

- SHARK's routing table don't have the subnet 192.168.1.0/24 and 192.168.1.1/32.
 - This happen because SHARK has not learned about the 192.168.1.0/24 network. So SHARK cannot route packets to this subnet.

f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

I will include the 192.168.1.0/24 subnet into the TIBURON's routing table by updating the RIP configuration.

g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON>enable
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#
TIBURON(config-router)#
```

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	<pre> C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=14ms TTL=126 Reply from 172.18.110.30: bytes=32 time=10ms TTL=126 Reply from 172.18.110.30: bytes=32 time=8ms TTL=126 Reply from 172.18.110.30: bytes=32 time=14ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 8ms, Maximum = 14ms, Average = 11ms </pre>
2	<pre> C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Request timed out. Reply from 172.18.110.62: bytes=32 time=10ms TTL=126 Reply from 172.18.110.62: bytes=32 time=10ms TTL=126 Reply from 172.18.110.62: bytes=32 time=12ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 10ms, Maximum = 12ms, Average = 10ms </pre>
4	<pre> C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Request timed out. Reply from 172.18.110.126: bytes=32 time=9ms TTL=126 Reply from 172.18.110.126: bytes=32 time=2ms TTL=126 Reply from 172.18.110.126: bytes=32 time=10ms TTL=126 Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 10ms, Average = 7ms </pre>

REFLECTION

What have you learned in this task?

First of all, I learn how to divide a network into smaller subnet. Then I learn how to assign IP address to both device in the network. Moreover, I learn that the RIP can constantly exchanges routing information with neighbouring routers so that all devices can connect with each other.